COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE031	**SECTION:**	CPE21S1
DATE PERFORMED:	Oct 10, 2026	**DATE SUBMITTED:**	Oct 10, 2026
NAME:	ANDREW DOLOSO JR. RALPH FLORENZ DUE?AS JAREL DAVE EGAR	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

DOLOSO

Task 1

Task:

Create one countplot for Product and one histogram for Profit.

Compare their patterns and interpret what they mean.

DOLOSO

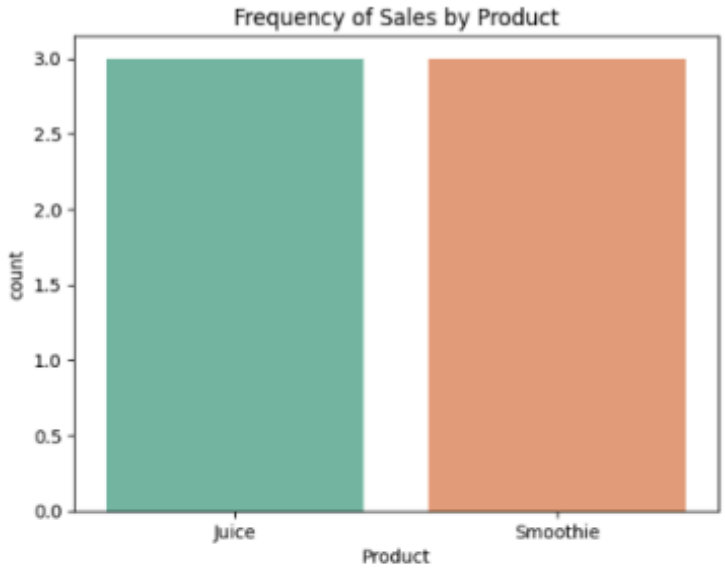
- The profit side has a different pattern though. It splits into lower amounts around twenty to thirty eight dollars and then higher ones from forty four to fifty with nothing in the middle. This happens despite the even split in how many of each product sold. Profitability changes quite a bit so other things like sizing or costs probably matter more than just the category.

```
#Code here
sns.countplot(x="Product", data=df, palette="Set2")
plt.title("Frequency of Sales by Product")
plt.show()

# Histogram for Profit
plt.hist(df["Profit"], bins=5, color="lightgreen", edgecolor="black")
plt.title("Distribution of Profit")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.show()
```

DOLOSO

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v
sns.countplot(x="Product", data=df, palette="Set2")
```





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Task 2

Task:
Interpret whether higher sales also mean higher profit.
Which region seems to perform best?

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- Base on my observation higher sales lead to higher profits, as is evident from the positive correlation in the scatter plot, which stands at 0.99. In the East, the region with the best performance, there is a total of 260 sales and 70 profits. This region also has the best individual performance with 170 sales and 50 profits.

Task 3

Task:
Describe what you observe in the missing data visualization.
Which variables need attention before analysis?

DOLOSO

- The scatter plot points between sales and profit. Higher sales seem to lead to higher profits most of the time. East does the best overall with the top numbers. Sales reach 170 and profit hits 50 from Smoothie sales. That part feels important but the other regions stay lower without clear reasons why. I think the correlation holds across the points though maybe not as tightly as it first looks.

Task 4

Task:
Add a moving average line (rolling mean) to smooth fluctuations.
Hint: Use ts_df["Sales"].rolling(window=3).mean().

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DOLOSO

```
#Code here
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Generate time-series example
dates = pd.date_range(start="2024-01-01", periods=10)
sales = np.random.randint(80, 200, 10)
ts_df = pd.DataFrame({"Date": dates, "Sales": sales})

# Moving average (3-day rolling mean)
ts_df["Moving_Avg"] = ts_df["Sales"].rolling(window=3).mean()

# Line chart for time-series with moving average
plt.plot(ts_df["Date"], ts_df["Sales"], marker="o", color="teal", label="Daily Sales")
plt.plot(ts_df["Date"], ts_df["Moving_Avg"], color="orange", linewidth=2.5,
         linestyle="--", label="3-Day Moving Average")
plt.title("Sales Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
dates = pd.date_range(start="2024-01-01", periods=10)
sales = np.random.randint(80, 200, 10)
ts_df = pd.DataFrame({"Date": dates, "Sales": sales})

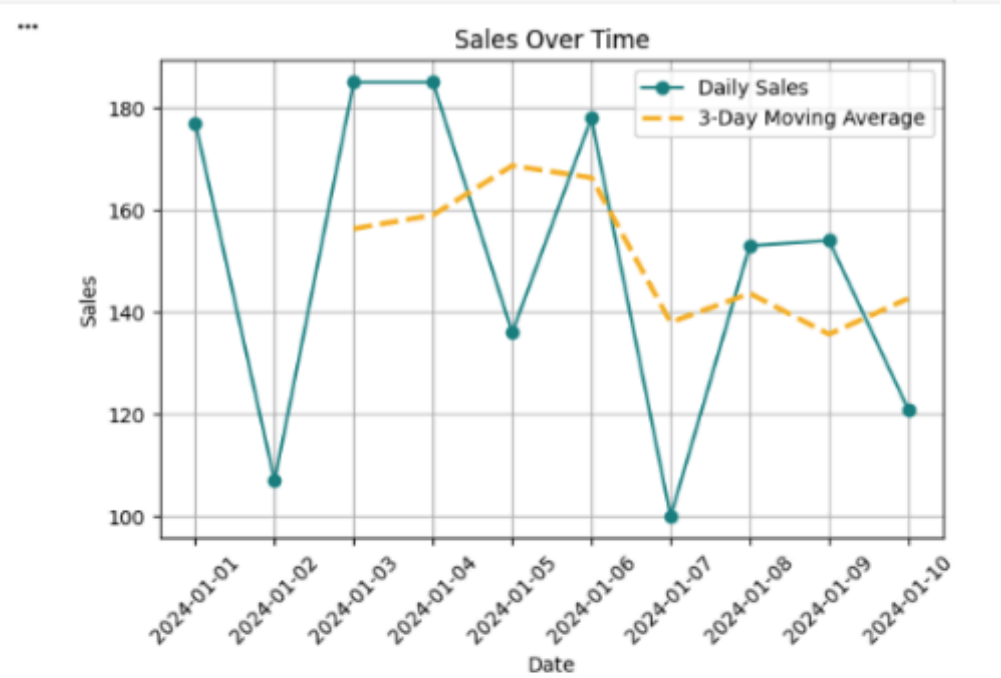
# Moving average (3-day rolling mean)
ts_df["Moving_Avg"] = ts_df["Sales"].rolling(window=3).mean()

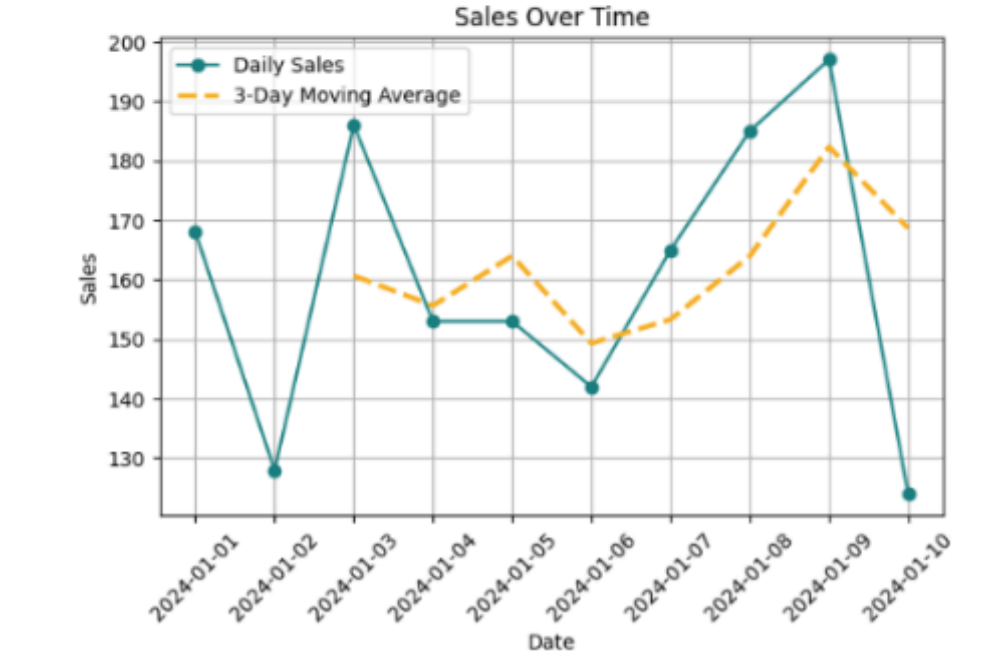
# Line chart for time-series with moving average
plt.plot(ts_df["Date"], ts_df["Sales"], marker="o", color="teal", label="Daily Sales")
plt.plot(ts_df["Date"], ts_df["Moving_Avg"], color="orange", linewidth=2.5,
         linestyle="--", label="3-Day Moving Average")
plt.title("Sales Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

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Task 5

Task:
Interpret which branch has the highest sales geographically.
How can such visualization help in business decision-making?

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- The map shows that South Branch has the highest sales with 150, while East Branch has the lowest with 90. This helps the business decide where to focus stocks and marketing. It can also help with expansion and delivery plans. However, other factors, such as cost and population, should also be considered.

DUENAS

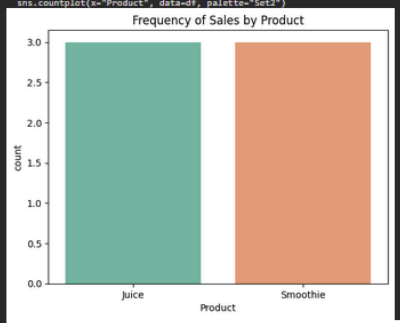
Task 1

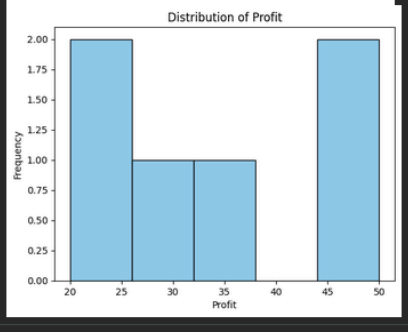
```
sns.countplot(x="Product", data=df, palette="Set2")
plt.title("Frequency of Sales by Product")
plt.show()

plt.hist(df["Profit"], bins=5, color="skyblue", edgecolor="black")
plt.title("Distribution of Profit")
plt.xlabel("Profit")
plt.ylabel("Frequency")
plt.show()
/tmp/ipykernel_1213/2722543527.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(x="Product", data=df, palette="Set2")
```





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Task 2

Section 2: Relationships Between Variables

```
# Scatterplot: Sales vs Profit
sns.scatterplot(x="Sales", y="Profit", hue="Region", data=df, palette="coolwarm", s=100)
plt.title("Relationship between Sales and Profit")
plt.show()
```




```
# Crosstabulation example
pd.crosstab(df["Region"], df["Product"], values=df["Sales"], aggfunc="mean").fillna(0)
```

Region	Juice	Smoothie
East	90.0	170.0
North	125.0	0.0
South	0.0	150.0
West	0.0	110.0

Task:
Interpret whether higher sales also mean higher profit.
Which region seems to perform best?

DUENAS

The higher the sales means higher profit because the graph shows an upward line. Among all areas, the East region performed the best overall. It reached both the highest single sale of 170 and the highest profit of 50.

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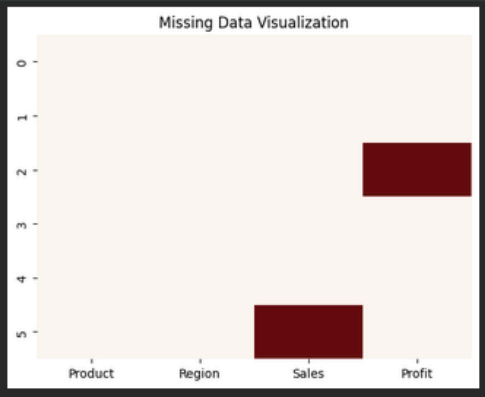
Task 3

Section 3: Analysis of Missing Data

```
# Introduce missing data
df_missing = df.copy()
df_missing.loc[2, "Profit"] = np.nan
df_missing.loc[5, "Sales"] = np.nan

# Visualize missing data
sns.heatmap(df_missing.isnull(), cbar=False, cmap="Reds")
plt.title("Missing Data Visualization")
plt.show()

# Display missing summary
df_missing.isnull().sum()
```



	e
Product	0
Region	0
Sales	1
Profit	1

dtype: int64

Task:
Describe what you observe in the missing data visualization.
Which variables need attention before analysis?

Duenas

The missing data visualization shows two dark red boxes, indicating missing values in row 2 and row 5. Specifically, there is one missing value in Sales and one missing value in Profit. Therefore, Sales and Profit are the two variables that need attention before running further analysis.

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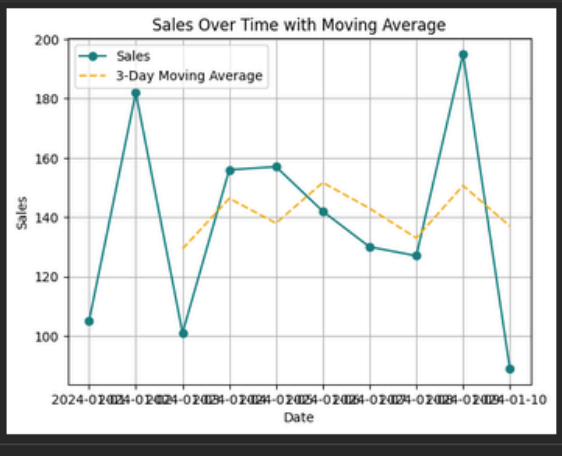
DUENAS

Task 4

Duenas

```
ts_df["Moving_Average"] = ts_df["Sales"].rolling(window=3).mean()

plt.plot(ts_df["Date"], ts_df["Sales"], marker="o", color="teal", label="Sales")
plt.plot(ts_df["Date"], ts_df["Moving_Average"], color="orange", linestyle="--", label="3-Day Moving Average")
plt.title("Sales Over Time with Moving Average")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.legend()
plt.grid(True)
plt.show()
```



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DUENAS

Task 5

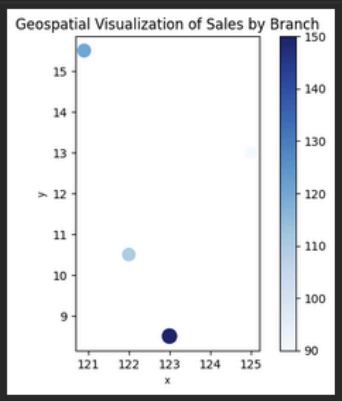
Section 5: Visualizing Geospatial Data

```
import geopandas as gpd
from shapely.geometry import Point

# Sample coordinates (latitude, longitude)
coords = {
    "Branch": ["North", "South", "East", "West"],
    "Latitude": [15.5, 8.5, 13.0, 10.5],
    "Longitude": [120.9, 123.0, 125.0, 122.0],
    "Sales": [120, 150, 90, 110]
}

geo_df = pd.DataFrame(coords)
geo_df["geometry"] = [Point(xy) for xy in zip(geo_df.Longitude, geo_df.Latitude)]
gdf = gpd.GeoDataFrame(geo_df, geometry="geometry")

# Plot simple map (Philippines outline optional if available)
gdf.plot(column="Sales", cmap="Blues", legend=True, markersize=geo_df["Sales"])
plt.title("Geospatial Visualization of Sales by Branch")
plt.show()
```



Task:
Interpret which branch has the highest sales geographically.
How can such visualization help in business decision-making?

The South branch has the highest sales geographically, represented by the darkest blue dot with 150 sales at the bottom of the map. This geospatial visualization helps business leaders quickly see which locations perform best to guide decisions on resource allocation and expansion. It also makes it easy to identify underperforming regions that may require extra marketing or operational support.

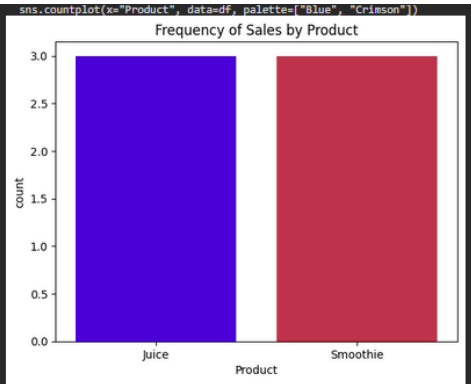
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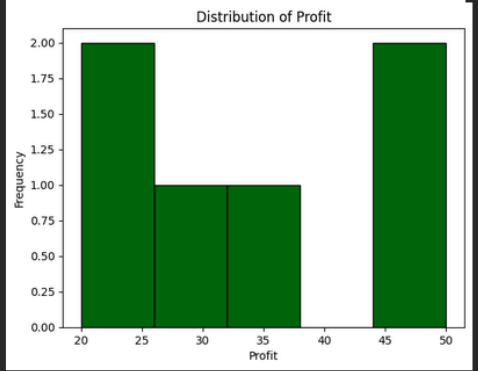
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EGAR

Task 1

```
sns.countplot(x="Product", data=df, palette=["Blue", "Crimson"])
```





As I observe the the two graphs the frequency of the two producsts are the same but in the 2nd graph, after 20 profit it hit the highest peak of 2.00 then after it hit the 27 to 36 profit values it went down to 1.00 then had zero frequency for profit 40 to 43 then it lastly hit 44 to 50 profit values with the frequency of 2.00.

Task 2

EGAR

I noticed that the region with the highest profit and sales is the region east. Because of this I noticed that if the sales are high that also means that the profit also increases.

Task 3

+ Code + Text

I noticed that there are 2 missing datas, which are in the column of sales and profit. When computing for sales and profit, it is crucial for the computation and if there are any missing data then it will provide false financial data.

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Task 4

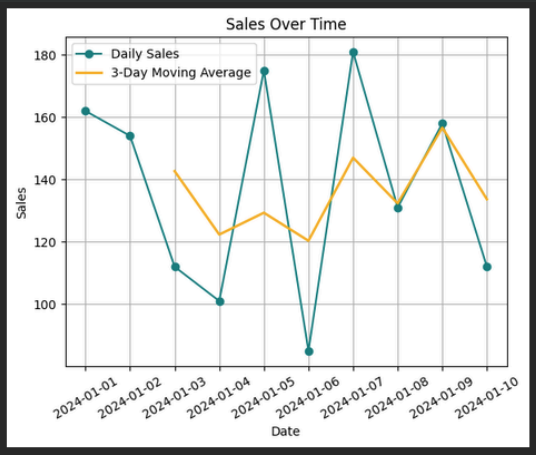
```
# Generate time-series example
dates = pd.date_range(start="2024-01-01", periods=10)
sales = np.random.randint(80, 200, 10)
ts_df = pd.DataFrame({"Date": dates, "Sales": sales})

# hint used
ts_df["Moving_Avg"] = ts_df["Sales"].rolling(window=3).mean()

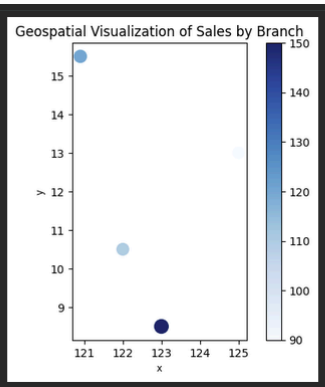
# Line chart for time-series
plt.plot(ts_df["Date"], ts_df["Sales"], marker="o", color="teal", label="Daily Sales")
plt.plot(ts_df["Date"], ts_df["Moving_Avg"], color="orange", linewidth=2, label="3-Day Moving Average")

plt.title("Sales Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.legend()


plt.grid(True) #gridl ines
plt.xticks(rotation=30) # tild by 30 degrees
plt.show()
```



Task 5



Task:
Interpret which branch has the highest sales geographically.
How can such visualization help in business decision-making?

Egar

The one with the highest sales geographically is the light Sea blue marker with y of 17 and heatmap of 118. this can significantly help busineses in many ways like finding the highest rated product and naming the heatmap for how many people liked it.

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SUPPLEMENTARY

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("/content/drive/MyDrive/most_subscribed_youtube_channels.csv")

# Convert text numbers like "222,000,000" into real numbers
for col in ["subscribers", "video views", "video count"]:
    df[col] = df[col].str.replace(",", "").astype(float)

df.head()
```

	rank	Youtuber	subscribers	video views	video count	category	started
0	1	T-Series	222000000.0	1.984591e+11	17317.0	Music	2006
1	2	YouTube Movies	154000000.0	0.000000e+00	0.0	Film & Animation	2015
2	3	Cocomelon - Nursery Rhymes	140000000.0	1.354813e+11	786.0	Education	2006
3	4	SET India	139000000.0	1.257643e+11	91271.0	Shows	2006
4	5	Music	116000000.0	0.000000e+00	0.0	NaN	2013

Next steps: New interactive sheet

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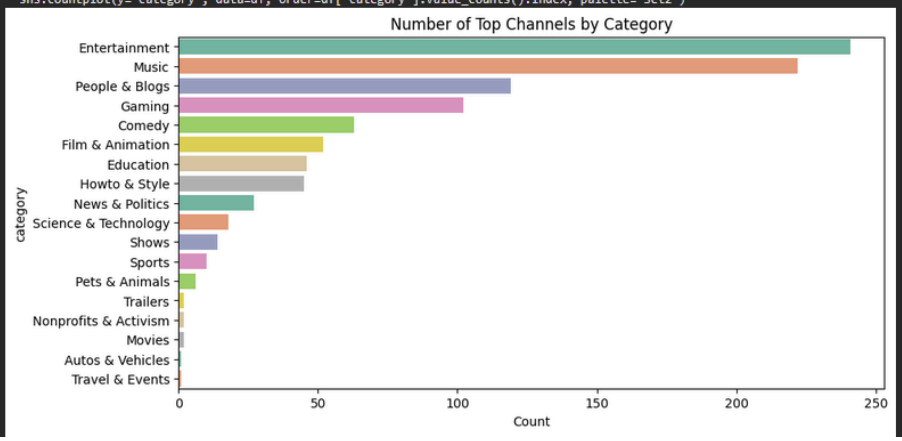
SUPPLEMENTARY

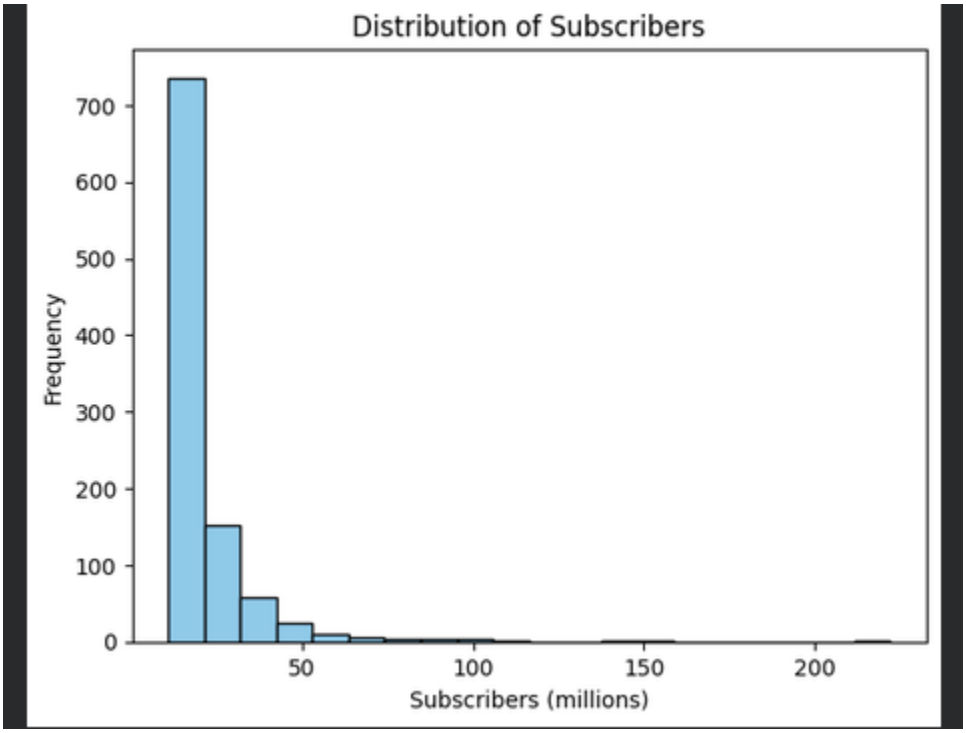
```
# UNIVARIATE
# The number of channels per category (Qualitative)
plt.figure(figsize=(10, 5))
sns.countplot(y="category", data=df, order=df["category"].value_counts().index, palette="Set2")
plt.title("Number of Top Channels by Category")
plt.xlabel("Count")
plt.show()

# The distribution of subscribers (Quantitative)
plt.hist(df["subscribers"] / 1_000_000, bins=20, color="skyblue", edgecolor="black")
plt.title("Distribution of Subscribers")
plt.xlabel("Subscribers (millions)")
plt.ylabel("Frequency")
plt.show()
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(y="category", data=df, order=df["category"].value_counts().index, palette="Set2")
```





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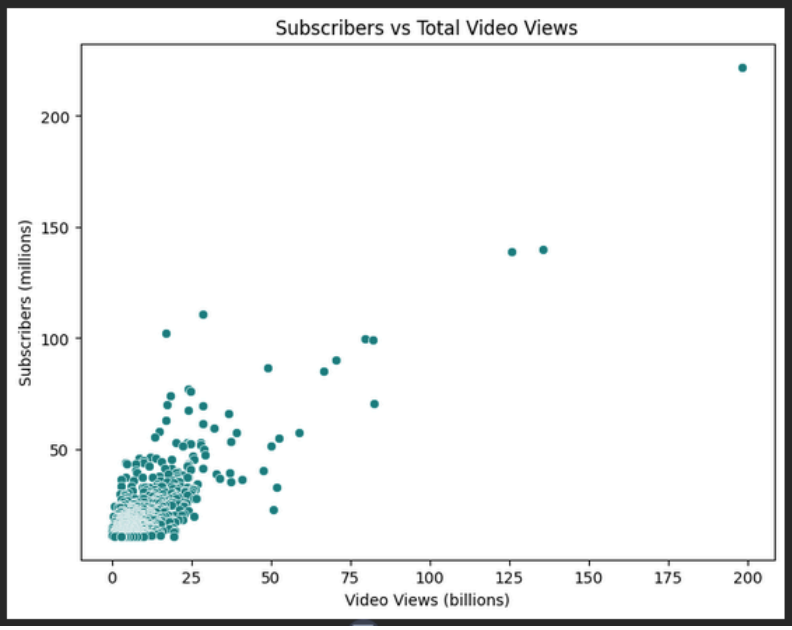
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SUPPLEMENTARY

```
#UNIVARIATE
# Remove the channels with 0 views since sometimes this are just music and gamplay videos with no views or other video with no views
df_views = df[df["video views"] > 0]

plt.figure(figsize=(8, 6))
sns.scatterplot(x=df_views["video views"] / 1e9, y=df_views["subscribers"] / 1e6, color="teal")
plt.title("Subscribers vs Total Video Views")
plt.xlabel("Video Views (billions)")
plt.ylabel("Subscribers (millions)")
plt.show()

# Correlation
df_views[["subscribers", "video views"]].corr()
```



	subscribers	video views
subscribers	1.000000	0.828314
video views	0.828314	1.000000

```
#BIVARIATE
# Missing data
df_missing = df.copy()
df_missing.loc[2, "subscribers"] = np.nan
df_missing.loc[5, "video views"] = np.nan

# Creating the heatmap for missing data
sns.heatmap(df_missing.isnull(), cbar=False, cmap="Reds", yticklabels=False)
plt.title("Missing Data Visualization")
plt.show()

# Display summary
df_missing.isnull().sum()
```



	0
rank	0
Youtuber	0
subscribers	0
video views	0
video count	0
category	27
started	0

dtype: int64

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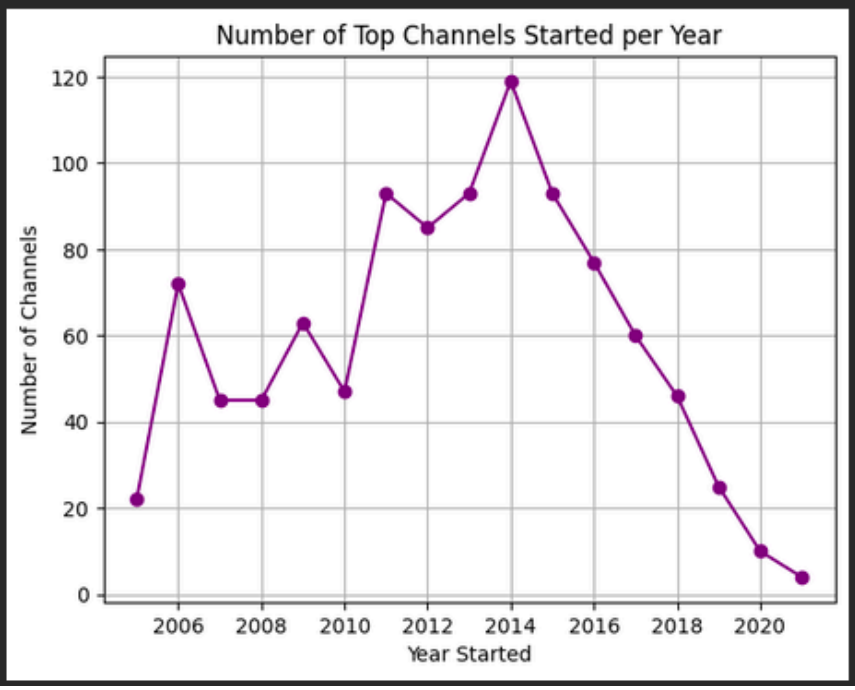
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SUPPLEMENTARY

```
#BIVARIATE
# Remove the 1970 entry since youtube was created in 2005
df_time = df[df["started"] >= 2005]

channels_per_year = df_time["started"].value_counts().sort_index()

plt.plot(channels_per_year.index, channels_per_year.values, marker="o", color="purple")
plt.title("Number of Top Channels Started per Year")
plt.xlabel("Year Started")
plt.ylabel("Number of Channels")
plt.grid(True)
plt.show()
```



EGAR - KEY INSIGHTS

1. I learned about Bivariete, Univariate, and Multivariate
2. I learned how to clean data properly
3. I learned how to use graphs to visualy represent the missing datas

CONCLUSION

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SUPPLEMENTARY

Egar

This activity has helped me understan how to properly clean datas by removing missing values or creating a synthetic value for the. We used many different graphs to visualize our data set. We also learned about Bivariete, Univariate, and Multivariate. Univariate is data with only one variable, like a countplot of categories or a histogram of subscribers. Bivariate is data with two variables, like a scatterplot that shows if more video views also means more subscribers. Multivariate is data with three or more variables, like a scatterplot with colors for each category. We also learned how to find missing data using a heatmap, and how to show changes over time using a line chart. In the end, I learned that using the right graph makes the data easier to understand and helps me see patterns, trends, and problems in the data.

Duenas

This HOA showed to clean data properly by removing missing values or replacing them with estimated numbers. We created various charts to explore our dataset clearly. We also studied univariate, bivariate, and multivariate analysis. Univariate looks at just one variable, such as a countplot of categories or a histogram of subscribers. Bivariate compares two variables, like using a scatterplot to check if more video views lead to more subscribers. Multivariate looks at three or more variables, such as adding different colors to a scatterplot for each group. Additionally, we learned how to spot missing values with a heatmap and track changes over time using a line chart. Overall, using the right graph makes information much simpler to read and helps us easily find patterns, trends, and errors in the data

Doloso

This process has really helped me understand how to cleanse my data by removing any missing values or introducing some synthetic value into it. We have used numerous gra for visualizing our dataset. Moreover, we have learned about Bivariete, Univariate, and Multivariate. Univariate is data with just one variable which could be shown as countplot categories or histogram for subscribers. Bivariate is a dataset with two variables shown scatterplot and representing correlation between video views and subscribers. Multivari data with three or more variables such as scatter plot with different colors for each cate We have learned how to detect missing data through heatmap and how to represent tim series data through line chart. In conclusion, I realized that proper use of a graph can m our data clearer and help us analyze its patterns, trends, and issues.

COLABS

DOLOSO

DOLOSO_Hands_On_Activity_9___Exploring_Data_Visually-1 (2).ipynb

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COLABS
DUENAS
Duenas_ Hands_On_Activity_9___Exploring_Data_Visually-1.ipynb
EGAR
EGAR_Hands_On_Activity_9___Exploring_Data_Visually-1(1).ipynb